

# Quadratic Completion of Admissible Spectral Pairs via Binary-Icosahedral Representation

O26 — Spectral Admissibility Sub-programme

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April 2026

## Abstract

We investigate the structural nature of the pair observable  $\sigma_{\text{pair}}(n) = \sigma_c(n) \sigma_{q-c}(n)$  arising in the Weil-block analysis of the Heisenberg Cayley graphs  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ . Building on the canonical construction of conjugate pairs enforced by the parity involution [1] and the normalisation invariance established in O17–O19 [2, 1, 3], we construct an explicit dictionary between conjugate Weil blocks and rank-one matrix coefficients in a representation space.

We show that, in the pre-saturation regime,  $\sigma_{\text{pair}}(n)$  has the same growth exponent as the Hilbert–Schmidt norm of an associated matrix trajectory, establishing a Level I identification (proved). We then formulate a hierarchy of stronger identifications: a quotient identification modulo normalisation (Level II), and a canonical representation-theoretic identification (Level III), conjectured to take place in an isotypic sector of the binary icosahedral group  $2I$  along the admissibility thread  $Q_8 \subset 2I \subset \text{SU}(2)$ .

We provide concrete falsifiability tests based on the effective dimension of the trajectory in  $\text{End}(V_\rho)$ , directly computable from O25 data [4]. A positive result would identify  $\sigma_{\text{pair}}$  as the restriction of a canonical Hermitian quadratic form and provide the first representation-theoretic interpretation of the admissible cascade exponent  $\beta^*$ . A negative result would still determine the correct ambient representation sector, refining the admissibility hierarchy without invalidating the dictionary.

## Contents

|          |                                                                                      |          |
|----------|--------------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Background and motivation</b>                                                     | <b>2</b> |
| <b>2</b> | <b>The admissible <math>2I</math> thread and the 600-cell</b>                        | <b>3</b> |
| <b>3</b> | <b>The dictionary: conjugate Weil blocks and isotypic sectors of <math>2I</math></b> | <b>3</b> |
| 3.1      | The Weil side: structure of $\sigma_{\text{pair}}^{\text{can}}$ . . . . .            | 3        |
| 3.2      | The $2I$ side: isotypic sectors and the Peter–Weyl quadratic form . . . . .          | 4        |
| 3.3      | The dictionary map . . . . .                                                         | 4        |
| 3.4      | Strength of the identification . . . . .                                             | 5        |

|          |                                                   |          |
|----------|---------------------------------------------------|----------|
| <b>4</b> | <b>The conjecture at three levels</b>             | <b>6</b> |
| 4.1      | Level I: Growth equivalence (proved)              | 6        |
| 4.2      | Level II: Quotient identification (structural)    | 6        |
| 4.3      | Level III: Canonical identification (conjectural) | 7        |
| 4.4      | Candidate sector                                  | 7        |
| 4.5      | Structural interpretation                         | 7        |
| <b>5</b> | <b>Discriminating criteria and falsifiability</b> | <b>7</b> |
| 5.1      | Test 1: Involution compatibility                  | 7        |
| 5.2      | Test 2: Gauge invariance                          | 8        |
| 5.3      | Test 3: Exponent stability                        | 8        |
| 5.4      | Test 4: Effective dimension of the trajectory     | 8        |
| 5.5      | Test 5: Universality across pairs                 | 9        |
| 5.6      | Decision table                                    | 9        |
| <b>6</b> | <b>Implications</b>                               | <b>9</b> |
| 6.1      | From bilinear observable to canonical norm        | 9        |
| 6.2      | Interpretation of the exponent $\beta^*$          | 9        |
| 6.3      | Selection of the admissible sector                | 10       |
| 6.4      | Role of non-injectivity                           | 10       |
| 6.5      | Outlook                                           | 10       |

# 1 Background and motivation

The spectral admissibility programme (O1–O25) studies the emergence of observable scaling laws from the bounded-flux dynamics of the Cosmochrony substrate  $\chi$  [5]. A central object is the Gram–Schmidt redundancy observable

$$\sigma_c(n) = \frac{\delta r_n}{|S_n|}, \quad (1)$$

defined on BFS shells  $S_n$  of Cayley graphs associated with  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ .

Early analyses based on scalar observables led to systematic discrepancies in the extracted exponent  $\delta$  (O11–O14 [6, 7, 8, 9]). This mismatch was resolved in O16 [10] by identifying the correct observable as the *conjugate pair observable*

$$\sigma_{\text{pair}}^{\text{can}}(n) = \sigma_c(n) \sigma_{q-c}(n), \quad (2)$$

which doubles the exponent and yields values in the admissible window  $\delta_{\text{pair}} \in [7.4, 10.6]$ .

The necessity of pairing is structural. O18 [1] shows that every fibre of the projection  $\Pi$  contains the parity involution  $\chi \mapsto -\chi$ , which induces the conjugation relation  $\rho_{q-c} = \overline{\rho_c}$  at the Weil level. O17–O19 [2, 1, 3] establish that the apparent asymmetry between  $c$  and  $q - c$  is a normalisation artefact, not a structural feature.

The present note addresses the following question:

Is the bilinear form  $\sigma_{\text{pair}}^{\text{can}}(n)$  an intrinsic quadratic object, or merely a product enforced by symmetry?

We show that  $\sigma_{\text{pair}}^{\text{can}}$  admits a natural interpretation as the pullback of a Hermitian quadratic form on a representation space, and formulate precise criteria under which this interpretation becomes canonical.

## 2 The admissible $2I$ thread and the 600-cell

The admissibility analysis of [11, 12] classifies the finite subgroups of  $SU(2)$  compatible with bounded-flux admissibility constraints, identifying the chain

$$Q_8 \subset 2I \subset SU(2), \quad (3)$$

where  $2I$  is the binary icosahedral group of order 120. Among the admissible families, the binary polyhedral groups  $2O$  and  $2I$  are distinguished by the isotropy of their second-moment tensor [12], making  $2I$  the maximal admissible finite subgroup of  $SU(2)$ .

The group  $2I$  admits irreducible representations of dimensions

$$d_\rho \in \{1, 2, 2, 3, 3, 4, 4, 5, 6\}, \quad (4)$$

and its matrix coefficient spaces  $\text{End}(V_\rho)$  have dimensions  $d_\rho^2$ , forming the canonical Peter–Weyl decomposition of  $L^2(2I)$ .

Geometrically,  $2I$  is the rotational symmetry group of the regular 600-cell in  $\mathbb{R}^4$ , whose 120 vertices correspond to the elements of  $2I$ . The Laplacian spectrum of this structure can be organised representation-theoretically, with multiplicity patterns consistent with the Peter–Weyl dimensions  $d_\rho^2$  of the irreps of  $2I$  [11]. This is not a coincidence imported from external geometry: the same representation-theoretic hierarchy governs both the spectral decomposition of  $2I$  and the organisation of admissible degrees of freedom along the thread  $Q_8 \subset 2I \subset SU(2)$ .

The question addressed in this note is whether the conjugate-pair observable  $\sigma_{\text{pair}}^{\text{can}}$  selects one of these isotypic sectors and realises its canonical quadratic form through the Gram–Schmidt dynamics.

## 3 The dictionary: conjugate Weil blocks and isotypic sectors of $2I$

This section establishes the explicit correspondence between the pair observable  $\sigma_{\text{pair}}^{\text{can}}$  on conjugate Weil blocks  $\{c, q - c\}$  and a natural Hermitian quadratic form on an isotypic sector of  $2I$ . The dictionary has three layers: the Weil side (§3.1), the  $2I$  side (§3.2), and the identification map (§3.3).

### 3.1 The Weil side: structure of $\sigma_{\text{pair}}^{\text{can}}$

Fix a prime  $q$  and a generic block  $c_{\text{block}} = (c_1, c_2, c_3)$  with  $c_i \neq 0$  and  $c_1 + c_2 + c_3 \not\equiv 0 \pmod{q}$ . The Gram–Schmidt redundancy at BFS shell  $n$  is

$$\sigma_c(n) = \frac{\delta r_n}{|S_n|}, \quad (5)$$

where  $\delta r_n$  counts the number of Weil fingerprint vectors in shell  $S_n$  linearly independent of the span built over shells  $0, \dots, n - 1$  [7]. The canonical pair observable is

$$\sigma_{\text{pair}}^{\text{can}}(n) = \sigma_c(n) \sigma_{q-c}(n). \quad (6)$$

The structural basis of this definition rests on two results from O17–O18. First, the conjugation identity  $\rho_{q-c} = \overline{\rho_c}$  together with the norm-invariance of Gram–Schmidt

orthogonalisation implies  $\delta_{q-c} = \delta_c$  for all admissible  $c$  [2]. Second, the proportionality  $\sigma_{q-c}(n) = r(c, q) \sigma_c(n)$ , with  $r(c, q) \in \{1, 2, 3, 4\}$  independent of  $n$ , holds to machine precision ( $|\sigma_{q-c}(n)/\sigma_c(n) - r(c, q)| < 10^{-14}$  for all tested pairs and shells) [10]. Critically, O17 [2] demonstrates that  $r(c, q)$  is a normalisation artefact of the pipeline determined by the initial supports  $(b_1, b_2)$  of each block sample, not an intrinsic invariant of the representation. The structurally robust quantity is therefore  $\delta_c = \delta_{q-c}$ , and the canonical form of the pair observable removes the artefact by working with the product  $\sigma_c(n) \sigma_{q-c}(n)$  rather than each factor in isolation.

*Remark 3.1* (Gauge-like nature of  $r(c, q)$ ). The factor  $r(c, q)$  plays the role of a gauge degree of freedom in the pipeline: it depends on sampling initial conditions, not on the representation. The matrix  $M_n$  defined in §3.3 must be constructed so that  $\|\widetilde{M}_n\|_{\text{HS}}^2$  is independent of this gauge choice, i.e. it absorbs  $r(c, q)$  into a normalisation convention rather than treating it as structural data.

### 3.2 The $2I$ side: isotypic sectors and the Peter–Weyl quadratic form

For each irrep  $\rho$  of  $2I$  with representation space  $V_\rho$  of dimension  $d_\rho$ , the Peter–Weyl theorem provides a canonical space of matrix coefficients:

$$\mathcal{H}_\rho = \{f : 2I \rightarrow \mathbb{C} \mid f(g) = \langle u, \rho(g)v \rangle, \ u, v \in V_\rho\} \simeq V_\rho \otimes \overline{V}_\rho \simeq \text{End}(V_\rho), \quad (7)$$

which has dimension  $d_\rho^2$ . This space carries a canonical Hermitian inner product, the Hilbert–Schmidt form:

$$\langle A, B \rangle_{\text{HS}} = \text{tr}(A^\dagger B), \quad \|A\|_{\text{HS}}^2 = \text{tr}(A^\dagger A), \quad A, B \in \text{End}(V_\rho). \quad (8)$$

For a rank-one element  $A = u \otimes \bar{v}$ , this reduces to

$$\|u \otimes \bar{v}\|_{\text{HS}}^2 = \|u\|^2 \cdot \|v\|^2. \quad (9)$$

This is the key structural fact: the Hilbert–Schmidt norm of a rank-one endomorphism is the product of the norms of its two factors.

### 3.3 The dictionary map

**Construction.** For each BFS shell  $n$  and each conjugate pair  $\{c, q - c\}$ , define the *un-normalised matrix coefficient*

$$\widetilde{M}_n := v_c^{(n)} \otimes \overline{v_{q-c}^{(n)}}, \quad (10)$$

where  $v_c^{(n)}$  is the vector whose  $s$ -th component is the residual norm  $\|r_s^{(n)}\|$  of the  $s$ -th Weil fingerprint vector in shell  $S_n$  after projection onto the Gram–Schmidt span built over shells  $0, \dots, n - 1$ , and similarly for  $v_{q-c}^{(n)}$ .

**Central identity.** By (9),

$$\|\widetilde{M}_n\|_{\text{HS}}^2 = \|v_c^{(n)}\|^2 \cdot \|v_{q-c}^{(n)}\|^2 \sim \sigma_c(n) \cdot \sigma_{q-c}(n) = \sigma_{\text{pair}}^{\text{can}}(n). \quad (11)$$

*Remark 3.2* (From pipeline quantity to structural norm). The proportionality  $\|v_c^{(n)}\|^2 \sim \sigma_c(n)$  in (11) is not an additional assumption; it follows from the definition of the Gram–Schmidt observable and a single normalisation choice.

With the convention above,  $\|v_c^{(n)}\|^2 = \sum_{s \in S_n} \|r_s^{(n)}\|^2$ , the total squared residual energy contributed by shell  $S_n$ . By construction,  $\sigma_c(n) = \delta r_n / |S_n|$ , where  $\delta r_n$  is the rank increment of shell  $S_n$ .

Under the canonical normalisation of the fingerprint vectors adopted in O12–O19 [7, 2, 1, 3], each fingerprint vector has unit initial norm. The proportionality

$$\|v_c^{(n)}\|^2 = \kappa \cdot |S_n| \cdot \sigma_c(n) + \varepsilon_n \quad (12)$$

holds with  $\kappa > 0$  a normalisation constant independent of  $n$ , and  $\varepsilon_n$  a remainder bounded by the contribution of vectors below the rank threshold. In the pre-saturation regime (the fitting window  $[n_0, n_1]$  used throughout O16–O25),  $\varepsilon_n / (|S_n| \cdot \sigma_c(n)) \rightarrow 0$  as the rank-threshold vectors become negligible relative to the total shell size.

Identity (11) therefore holds in the pre-saturation regime with a proportionality constant  $\kappa^2 |S_n|^2$  absorbed into the pipeline normalisation. The power-law exponent  $\delta_{\text{pair}}$ , extracted from the log-log regression of  $\sigma_{\text{pair}}^{\text{can}}(n)$ , is unaffected by this constant factor.

### Three verification criteria.

- (i) **Involution compatibility.** The parity involution  $c \leftrightarrow q - c$  corresponds, via the conjugation identity  $\rho_{q-c} = \overline{\rho_c}$ , to complex conjugation  $v \mapsto \bar{v}$  on the representation space. This is precisely the anti-linear involution that exchanges  $u \otimes \bar{v}$  and  $v \otimes \bar{u}$  in  $\text{End}(V_\rho)$ , i.e. Hermitian adjunction  $A \mapsto A^\dagger$ . The pair observable  $\sigma_{\text{pair}}^{\text{can}}(n) = \sigma_c(n) \sigma_{q-c}(n)$  is symmetric under this exchange, consistent with  $\|A\|_{\text{HS}}^2 = \|A^\dagger\|_{\text{HS}}^2$ . Involution compatibility holds.
- (ii) **Gauge invariance.** The observable  $\sigma_{\text{pair}}^{\text{can}}(n)$  is invariant under rescaling  $v_c^{(n)} \mapsto \lambda v_c^{(n)}$ ,  $v_{q-c}^{(n)} \mapsto \lambda^{-1} v_{q-c}^{(n)}$  with  $\lambda > 0$ , because  $\sigma_c(n)$  scales as  $\lambda^2$  and  $\sigma_{q-c}(n)$  as  $\lambda^{-2}$ , leaving the product unchanged. Correspondingly,  $\|\widetilde{M}_n\|_{\text{HS}}^2$  is invariant under  $(u, v) \mapsto (\lambda u, \lambda^{-1} v)$  in the rank-one element  $u \otimes \bar{v}$ . Gauge invariance holds.
- (iii)  **$\delta_{\text{pair}}$  stability.** The exponent  $\delta_{\text{pair}}$  is extracted from the log-log regression of  $\sigma_{\text{pair}}^{\text{can}}(n)$  over the fitting window. By (11), this is the same as the log-log regression of  $\|\widetilde{M}_n\|_{\text{HS}}^2$ . The verticality theorem of O24 [4] establishes that  $\delta_{\text{pair}}$  is stable under fibre perturbations; this translates, via the dictionary, into the statement that  $\|\widetilde{M}_n\|_{\text{HS}}^2$  grows with a well-defined power-law exponent independent of the choice of conjugate pair.  $\delta_{\text{pair}}$  stability holds.

## 3.4 Strength of the identification

Identity (11) establishes the correspondence at *intermediate strength* in the sense of §4:  $\sigma_{\text{pair}}^{\text{can}}(n)$  is proportional to  $\|\widetilde{M}_n\|_{\text{HS}}^2$ , with a proportionality constant that is pipeline-normalisation-dependent and hence not intrinsic.

The *strong* identification requires the proportionality constant to be canonically fixed by the  $2I$ -representation theory, i.e. that  $\widetilde{M}_n$  takes values in a specific isotypic component  $\mathcal{H}_\rho$  for a representation  $\rho$  selected by the admissibility thread  $Q_8 \subset 2I \subset \text{SU}(2)$ . The admissibility hierarchy of [11] selects the spin- $\frac{1}{2}$  sector (dimension  $d_\rho = 2$ , Laplacian

eigenvalue  $\lambda_{1/2} = 18$ ) as the dominant admissible sector on the  $2I$  Cayley graph. If  $v_c^{(n)} \in V_\rho$  for this sector, then  $\widetilde{M}_n \in \text{End}(V_\rho)$  has  $d_\rho^2 = 4$  components. This identification is at present a conjecture, formulated precisely in §4.

## 4 The conjecture at three levels

Building on the dictionary of §3, we formulate a graduated identification between the pair observable  $\sigma_{\text{pair}}^{\text{can}}(n)$  and the Hilbert–Schmidt quadratic form on an isotypic sector of  $2I$ . The three levels differ in the strength of the claim: growth equivalence (proved), quotient identification (structural), and canonical identification (conjectural).

### 4.1 Level I: Growth equivalence (proved)

**Proposition 4.1** (Level I). *In the pre-saturation regime  $n \in [n_0, n_1]$ , the pair observable and the Hilbert–Schmidt norm of  $\widetilde{M}_n$  have identical power-law growth exponents:*

$$\delta_{\text{pair}} = \delta_{\text{HS}}, \quad (13)$$

where  $\delta_{\text{HS}}$  is defined by the log-log regression of  $\|\widetilde{M}_n\|_{\text{HS}}^2$  over the same fitting window.

*Proof.* By (11) and (12),  $\sigma_{\text{pair}}^{\text{can}}(n) = \kappa^2 |S_n|^2 \|\widetilde{M}_n\|_{\text{HS}}^2 + O(\varepsilon_n)$ , where the prefactor  $\kappa^2 |S_n|^2$  is independent of the conjugate pair. Taking log-log regression over  $[n_0, n_1]$  and noting that  $|S_n|$  grows as a fixed power of  $n$  (determined by the BFS geometry of  $G_q$ , independent of  $c$ ), the regressions of  $\log \sigma_{\text{pair}}^{\text{can}}$  and  $\log \|\widetilde{M}_n\|_{\text{HS}}^2$  differ only in their intercepts. Hence  $\delta_{\text{pair}} = \delta_{\text{HS}}$ .  $\square$

*Remark 4.2.* Level I is not a conjecture but a direct consequence of the dictionary construction of §3. It is stated separately to clarify what is proved and what remains conjectural at Levels II and III.

### 4.2 Level II: Quotient identification (structural)

The proportionality factor  $\kappa^2 |S_n|^2$  has two sources: the global normalisation convention of the fingerprint vectors (absorbed in  $\kappa$ ) and the shell-size scaling (absorbed in  $|S_n|^2$ ). Both are pipeline conventions rather than representation-theoretic data. Define the *normalisation equivalence*

$$A \approx B \iff A = C \cdot B, \quad (14)$$

where  $C > 0$  depends only on pipeline conventions and the BFS geometry of  $G_q$ , but not on the conjugate pair  $\{c, q - c\}$  or the shell index  $n$ .

**Conjecture 4.3** (Level II). *In the normalisation equivalence class defined above,*

$$\sigma_{\text{pair}}^{\text{can}}(n) \approx \|\widetilde{M}_n\|_{\text{HS}}^2. \quad (15)$$

*Equivalently,  $\sigma_{\text{pair}}^{\text{can}}(n)$  is the pullback of the Hilbert–Schmidt quadratic form on  $\text{End}(V_\rho)$  under the dictionary map (10), modulo a constant depending only on pipeline conventions.*

Level II would follow from Level I together with a proof that the proportionality constant  $\kappa^2 |S_n|^2$  does not depend on  $n$  in a way that could alter the pair-level structure. The  $n$ -dependence of  $|S_n|^2$  is BFS-geometric and uniform across pairs, so the primary open point is the  $n$ -independence of  $\kappa$ .

### 4.3 Level III: Canonical identification (conjectural)

The strong form requires the proportionality constant to be fixed canonically by the representation theory of  $2I$ , not merely absorbed into a normalisation convention.

**Conjecture 4.4** (Level III). *There exists an irreducible representation  $\rho$  of  $2I$ , selected by the admissibility thread  $Q_8 \subset 2I \subset \text{SU}(2)$ , such that for all admissible pairs  $\{c, q - c\}$  and all  $n$  in the pre-saturation regime:*

- (i) *the residual vector  $v_c^{(n)}$  lies in  $V_\rho$  under a canonical embedding determined by  $\rho$  and  $q$ ;*
- (ii) *the un-normalised matrix coefficient  $\widetilde{M}_n = v_c^{(n)} \otimes \overline{v_{q-c}^{(n)}}$  takes values in  $\text{End}(V_\rho)$ ;*
- (iii) *the proportionality constant between  $\sigma_{\text{pair}}^{\text{can}}(n)$  and  $\|\widetilde{M}_n\|_{\text{HS}}^2$  equals  $d_\rho^{-2}$ , the inverse squared dimension of  $\rho$ , up to the universal BFS-geometric factor  $|S_n|^2$ .*

### 4.4 Candidate sector

The admissibility hierarchy of [11] selects the spin- $\frac{1}{2}$  sector of  $2I$  as the minimal non-abelian admissible sector: dimension  $d_\rho = 2$ , Laplacian eigenvalue  $\lambda_{1/2} = 18$  on the standard  $2I$  Cayley graph. Conjecture 4.4(iii) would then read: the proportionality constant is  $d_\rho^{-2} = \frac{1}{4}$ , and  $\widetilde{M}_n$  takes values in a 4-dimensional space  $\text{End}(V_\rho) \simeq \mathbb{C}^{2 \times 2}$ .

This prediction is falsifiable: a direct computation of  $\text{rank}(\mathcal{C}_c)$  from the O25 pipeline data (see §5.4) determines whether the trajectory of  $\widetilde{M}_n$  is consistent with a 4-dimensional ambient space or requires a larger sector.

### 4.5 Structural interpretation

At Level I,  $\sigma_{\text{pair}}$  and the Hilbert–Schmidt norm share a growth exponent: the identification is asymptotic. At Level II,  $\sigma_{\text{pair}}$  is the Hilbert–Schmidt norm up to a normalisation: the identification is structural modulo convention. At Level III,  $\sigma_{\text{pair}}$  is the restriction of a canonical Hermitian quadratic form on a representation space selected by the admissibility thread: the pair observable is no longer a derived bilinear product, but a norm.

## 5 Discriminating criteria and falsifiability

The identification proposed in §4 leads to concrete, computable tests distinguishing the three levels. All tests require no modification of the O25 pipeline and can be evaluated directly from the stored Gram–Schmidt data.

### 5.1 Test 1: Involution compatibility

At the scalar level, the conjugation identity  $\rho_{q-c} = \overline{\rho_c}$  implies  $\delta_{q-c} = \delta_c$  [2]. At the matrix level, the dictionary maps  $c \leftrightarrow q - c$  to  $\widetilde{M}_n \leftrightarrow \widetilde{M}_n^\dagger$  in  $\text{End}(V_\rho)$ . This is an independent empirical prediction: if the identification is structural, the Frobenius distance

$$\Delta_n(c) := \|\widetilde{M}_n(c)^\dagger - \widetilde{M}_n(q - c)\|_{\text{HS}} \quad (16)$$

must vanish up to numerical precision for all  $n \in [n_0, n_1]$ .

**Criterion 5.1** (Involution).  $\Delta_n(c) < \varepsilon_{\text{mach}}$  for all tested pairs and shells.

A non-zero  $\Delta_n(c)$  that is nonetheless small would indicate that the identification holds only approximately, corresponding to Level II. A systematic non-zero value would falsify Levels II and III simultaneously.

## 5.2 Test 2: Gauge invariance

The factor  $r(c, q)$  is a normalisation artefact of the pipeline [2]. The dictionary absorbs it by working with  $\widetilde{M}_n$  rather than with individual normalised vectors.

**Criterion 5.2** (Gauge invariance). Under the rescaling  $v_c^{(n)} \mapsto \lambda v_c^{(n)}$ ,  $v_{q-c}^{(n)} \mapsto \lambda^{-1} v_{q-c}^{(n)}$  for  $\lambda > 0$ ,  $\|\widetilde{M}_n\|_{\text{HS}}^2$  is invariant.

This follows from (9) by direct computation:  $\|\lambda u \otimes \overline{\lambda^{-1} v}\|_{\text{HS}}^2 = \lambda^2 \|u\|^2 \cdot \lambda^{-2} \|v\|^2 = \|u\|^2 \cdot \|v\|^2$ . Test 2 is a consistency check on the construction, expected to hold identically.

## 5.3 Test 3: Exponent stability

Level I (Proposition 4.1) predicts  $\delta_{\text{pair}} = \delta_{\text{HS}}$ .

**Criterion 5.3** (Exponent stability). For each prime  $q \in \{29, 61, 101, 151\}$ , compute

$$\delta_{\text{HS}}(q) = \text{slope}(\log \|\widetilde{M}_n\|_{\text{HS}}^2 \text{ vs } \log(n+1)) \quad (17)$$

over the window  $[n_0, n_1]$  used in O25 (Table 1 therein; DOI pending deposit). The difference  $|\delta_{\text{HS}}(q) - \delta_{\text{pair}}(q)|$  must lie within the regression standard error reported in O25.

This test is the numerical anchor for the entire construction.

## 5.4 Test 4: Effective dimension of the trajectory

This is the central falsifiability test for Level III. For each conjugate pair  $\{c, q-c\}$ , construct the empirical covariance operator

$$\mathcal{C}_c = \frac{1}{N} \sum_{n=n_0}^{n_1} \text{vec}(\widetilde{M}_n) \text{vec}(\widetilde{M}_n)^\dagger, \quad N = n_1 - n_0 + 1, \quad (18)$$

where  $\text{vec}$  flattens  $\widetilde{M}_n$  into a column vector.

**Criterion 5.4** (Effective dimension). Let  $r_{\text{eff}}$  be the numerical rank of  $\mathcal{C}_c$  (number of singular values above threshold  $10^{-10} \cdot \sigma_1(\mathcal{C}_c)$ ).

- If  $r_{\text{eff}} = 4$  for all pairs: consistent with the spin- $\frac{1}{2}$  sector candidate ( $d_\rho = 2$ ,  $\dim \text{End}(V_\rho) = 4$ ).
- If  $r_{\text{eff}} = d_\rho^2$  for some other fixed  $d_\rho$ : Level III holds for a different irrep of  $2I$ ; the candidate sector must be revised.
- If  $r_{\text{eff}}$  varies across pairs or primes: Level III fails; the identification holds at most at Level II.

*Remark 5.5.* The matrix  $\widetilde{M}_n$  is reconstructed from the stored vectors  $v_c^{(n)}$  and  $v_{q-c}^{(n)}$  by outer product. The covariance (18) and its SVD scale as  $O(Nd^4)$  with  $d$  the effective dimension; for  $d = 2$  this is negligible. No additional BFS computation or Weil decomposition is required.



## 5.5 Test 5: Universality across pairs

Level III requires that all admissible pairs for a given  $q$  probe the same representation sector.

**Criterion 5.6** (Universality). The singular value spectra of  $\mathcal{C}_c$  and  $\mathcal{C}_{c'}$  must agree up to numerical fluctuations for all admissible pairs  $c, c'$  at the same prime  $q$ .

A pair-dependent spectrum would indicate that different pairs project onto different subspaces of  $\text{End}(V_\rho)$ , which is incompatible with a canonical identification. The inter-pair concentration of  $\delta_{\text{pair}}$  established in O25 (coefficient of variation dropping from 6.1% to 1.5% over  $q \in \{29, 61, 101, 151\}$ ) provides indirect support for this test.

## 5.6 Decision table

| Outcome                                               | Interpretation                                    |
|-------------------------------------------------------|---------------------------------------------------|
| Tests 1–3 fail                                        | Dictionary construction fails; §3 must be revised |
| Tests 1–3 pass, $r_{\text{eff}}$ variable             | Level I only                                      |
| Tests 1–3 pass, $r_{\text{eff}}$ fixed $\neq 4$       | Level II; wrong sector candidate                  |
| Tests 1–4 pass ( $r_{\text{eff}} = 4$ ), Test 5 fails | Level II with spin- $\frac{1}{2}$ sector          |
| Tests 1–5 pass                                        | Level III; Conjecture 4.4 confirmed               |

The row “Tests 1–3 pass,  $r_{\text{eff}}$  fixed  $\neq 4$ ” is structurally informative even as a negative result: it identifies the correct representation sector without confirming the spin- $\frac{1}{2}$  candidate, pointing to a specific revision of the admissibility hierarchy argument.

# 6 Implications

## 6.1 From bilinear observable to canonical norm

At Level III,  $\sigma_{\text{pair}}$  is the restriction of a canonical Hermitian quadratic form on  $\text{End}(V_\rho)$  selected by the admissibility thread  $Q_8 \subset 2I \subset \text{SU}(2)$ . The emergence of the pair observable is interpreted as a *quadratic completion*: the passage from a linear object  $v_c^{(n)} \in V_\rho$  to a rank-one operator  $\widetilde{M}_n \in \text{End}(V_\rho)$  endowed with its natural norm.

## 6.2 Interpretation of the exponent $\beta^*$

The transfer chain  $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$  was established unconditionally in O24 [4], where  $\beta^*$  is defined via

$$\beta^* \approx \frac{1}{\delta_{\text{pair}} + \frac{1}{2}}. \quad (19)$$

Under Level III,  $\delta_{\text{pair}}$  characterises the growth of a Hilbert–Schmidt norm along a trajectory in  $\text{End}(V_\rho)$ , and  $\beta^*$  acquires a direct representation-theoretic meaning: it is the effective scaling exponent of norm growth in the minimal admissible non-abelian sector. This provides a structural interpretation of the phenomenological window  $\beta^* \in (0.09, 0.13)$  [4] as a constraint on admissible trajectories in representation space, rather than as a purely numerical fit.

### 6.3 Selection of the admissible sector

The spin- $\frac{1}{2}$  sector emerges as a candidate not from external physical input, but from internal admissibility constraints [11]. Test 4 of §5.4 provides a direct verification of this selection. If confirmed,  $\text{End}(V_\rho) \simeq \mathbb{C}^{2 \times 2}$  is the minimal ambient space for admissible spectral dynamics. If a different fixed dimension is observed, the admissibility hierarchy must be revised accordingly, but the dictionary of §3 remains valid.

### 6.4 Role of non-injectivity

The construction makes explicit how non-injectivity of the projection  $\Pi : \chi \rightarrow \mathcal{O}$  induces a quadratic structure at the observable level. The pairing  $\{c, q - c\}$  is not an arbitrary symmetrisation, but the minimal realisation of the parity involution  $\chi \mapsto -\chi$  derived in O18 [1] from the Born–Infeld action. The Hermitian structure of  $\text{End}(V_\rho)$  is therefore a projection-induced structure: it does not exist at the level of the substrate  $\chi$ , but emerges from the identification of conjugate fibres under  $\Pi$ .

### 6.5 Outlook

The results of this note isolate a concrete bridge between the Weil representation of  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ , the admissibility thread  $Q_8 \subset 2I \subset \text{SU}(2)$ , and the observable exponent  $\delta_{\text{pair}}$ . The tests of §5 can be executed on O25 data without additional computation. A positive result at Level III would provide the first representation-theoretic realisation of the admissible cascade exponent. A negative result with a fixed effective dimension  $r_{\text{eff}} \neq 4$  would identify the correct ambient sector, refining the admissibility hierarchy without invalidating the dictionary.

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